Electronics – diodes, transistors and amplifiers

Where circuit theory deals with linear components, electronics deals with the non-linear ones: the diode, which conducts only one way, and the transistor, which lets a small signal control a large one. At Sussex there were two electronics courses for physicists, Electronics I and II, with theory and laboratory work. This page goes through the basic ideas with an interactive rectifier and a transistor amplifier.

> University of Sussex, Electronics I and II (1975–78): Two courses in the physics syllabus. The physics of semiconductors connects with solid-state physics; it was later followed up by circuit theory and digital electronics at DTH.

Semiconductors

In a semiconductor such as silicon the valence band is full and the conduction band empty at absolute zero, but the band gap is only 1.1 eV, so at room temperature a few electrons are lifted up, leaving holes that also conduct (see solid-state physics). Doping the crystal with small amounts of phosphorus (five valence electrons) gives an n-type semiconductor with free electrons; with boron (three), a p-type with free holes. The concentrations obey the law of mass action $np = n_i^2$, where $n_i \approx 10^{10}$ cm^{-3} for silicon at 300 K.

The diode

When p- and n-type silicon are placed together, electrons and holes diffuse across the boundary and recombine. What remains is a depletion layer of fixed ions and an electric field – a built-in voltage of about 0.7 V. A forward voltage lowers the barrier exponentially; in reverse only a small saturation current flows. Shockley's diode equation is

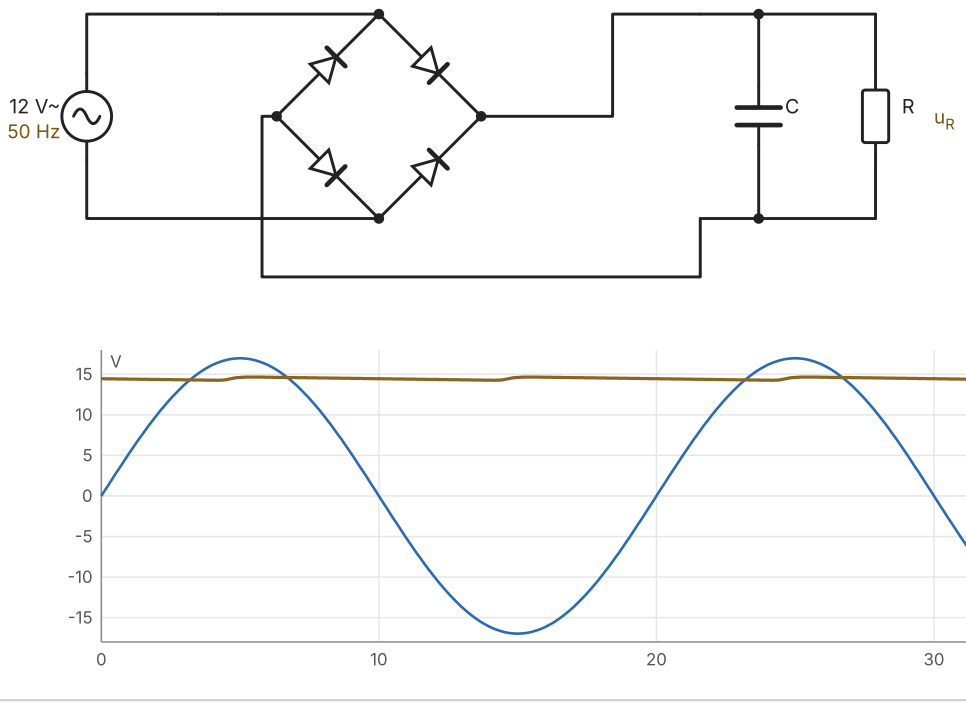
$$I = I_S\left(e^{U/nU_T} - 1\right), \qquad U_T = \frac{k_B T}{e} \approx 25.9\ \text{mV}$$

where n lies between 1 and 2. The current increases tenfold for every 60 mV, so in practice a silicon diode "opens" at about 0.6–0.7 V. The Zener diode is built to break down in reverse at a well-defined voltage and is used as a voltage reference. The light-emitting diode (LED) emits light when electrons and holes recombine in a semiconductor with a direct band gap, such as gallium arsenide.

Rectification and smoothing

A power supply must make a low DC voltage from the AC mains: a transformer steps the voltage down, diodes rectify it and a capacitor smooths it. With full-wave rectification in a bridge both half-cycles are turned positive. While the voltage is high the capacitor charges to the peak value; the rest of the time it discharges through the load. The remaining ripple is roughly

$$\Delta U \approx \frac{I}{f_r C}, \qquad f_r = \begin{cases} f & \text{half-wave} \\ 2f & \text{full-wave} \end{cases}$$



The transistor

John Bardeen, Walter Brattain and William Shockley invented the transistor at Bell Labs in 1947–48 and received the Nobel Prize in 1956. A bipolar transistor (BJT) is two pn junctions back to back: emitter, base and collector. A small base current I_B controls a much larger collector current $I_C = \beta I_B$, where the current gain β is typically 100–300. More precisely the collector current is set by the base–emitter voltage, $I_C = I_S e^{U_{BE}/U_T}$, and the transconductance $g_m = I_C/U_T$ is the most important parameter for amplification: 40 mA/V per mA of collector current.

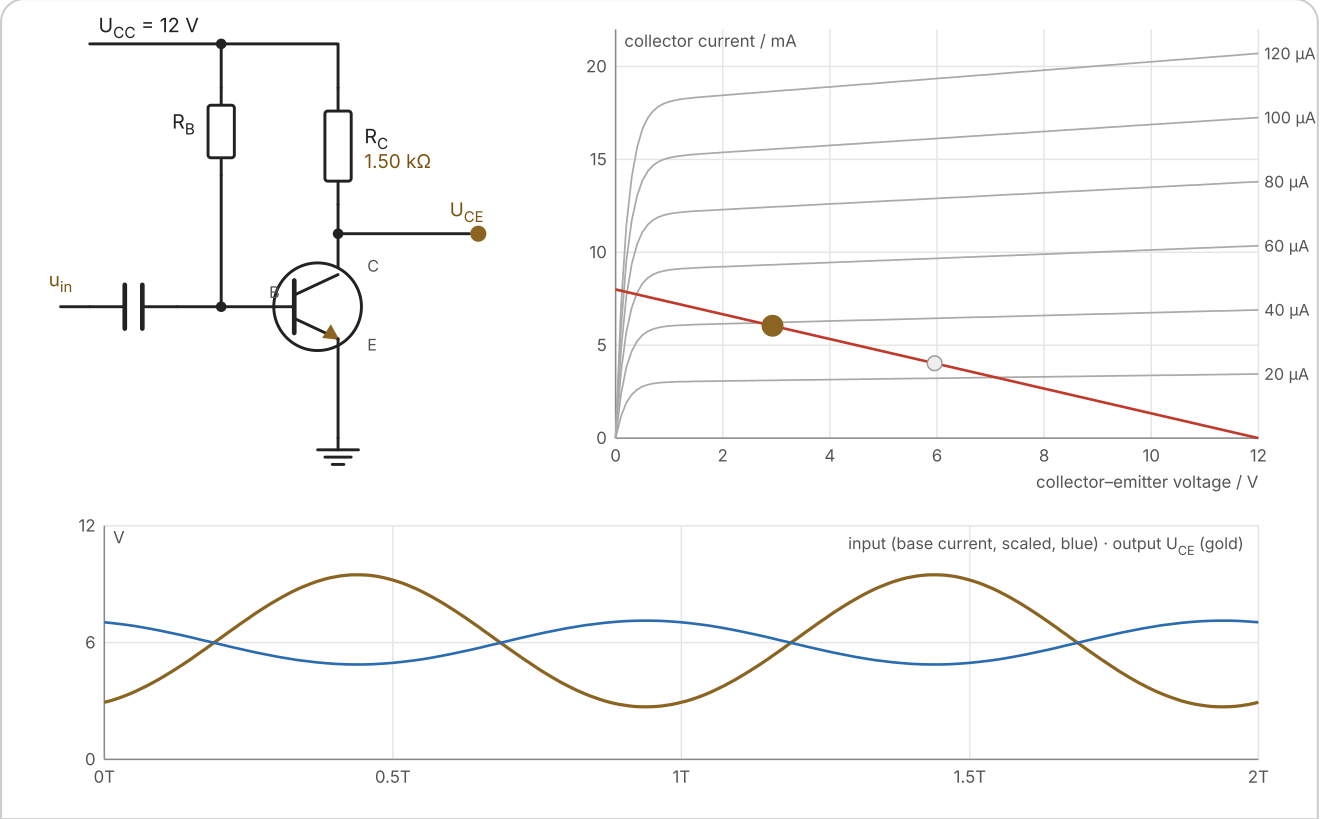
The amplifier

In a common-emitter amplifier the transistor is placed in series with a collector resistor R_C from the supply. The collector voltage is $U_{CE} = U_{CC} - I_C R_C$ – a straight line on the transistor

characteristics, the *load line*. The base bias is chosen so that the operating point lies in the middle of the line; then the signal can swing equally up and down. The voltage gain is roughly

$$A_u \approx -g_m R_C = -\frac{I_C R_C}{U_T}$$

and the sign means the output is in antiphase. If the operating point is too high or too low, the signal clips: the transistor saturates (fully on) or cuts off (off). An emitter resistor stabilises the operating point against temperature changes – a first example of negative feedback.



The field-effect transistor

In a field-effect transistor (FET) the current in a channel is controlled by an electric field from an insulated control electrode (gate) – with almost no gate current. The MOSFET, in which the gate is separated from the channel by a thin layer of silicon dioxide, was demonstrated by Mohamed Atalla and Dawon Kahng in 1959 and became the basis of integrated circuits. In the 1970s FETs were known above all as amplifiers with a very high input impedance – ideal for condenser microphones and pH electrodes, for example.

Feedback

Harold Black of Bell Labs had the idea of negative feedback in 1927 on the ferry across the Hudson: if a fraction β of the output is taken and subtracted from the input, the gain becomes

$$A_f = \frac{A}{1 + A\beta} \approx \frac{1}{\beta} \quad (A\beta \gg 1)$$

It becomes smaller, but is now almost independent of the spread and temperature of the transistors, and distortion and noise from the amplifier are reduced by the factor $1 + A\beta$. It is the principle behind the operational amplifier (see circuit theory) and all modern measuring electronics. If the feedback becomes positive at some frequency, the circuit oscillates – and that is the basis of oscillators.

Oscillators and noise

An oscillator is an amplifier with a frequency-selective network in the feedback. Barkhausen's criterion says that the loop gain must be exactly 1 with a phase shift of 0° (or 360°). The Wien-bridge oscillator, the basis of William Hewlett and David Packard's first product in 1939, uses an RC network that gives exactly 0° at $f = 1/(2\pi RC)$. Crystal oscillators use the sharp mechanical resonance of a quartz crystal and are accurate to parts per million.

All electronics has noise. A resistor produces thermal noise (Johnson–Nyquist, 1928) with the voltage $\overline{u^2} = 4k_B T R\,\Delta f$ – about 4 nV/√Hz for 1 kΩ at room temperature – and currents of individual charges give shot noise $\overline{i^2} = 2eI\,\Delta f$. Noise sets the lower limit on how weak a signal can be measured; that is why sensitive amplifiers for radio astronomy are cooled with liquid helium.

Since 1978

- **From discrete components to chips:** In the 1970s amplifiers were built from individual transistors; today almost all electronics is integrated circuits, and a mobile phone has more than ten billion transistors.
- **CMOS everywhere:** The MOSFET has almost entirely replaced the bipolar transistor in digital electronics, while bipolar and SiGe transistors are still used in fast analogue and radio-frequency circuits.
- **Power electronics:** Silicon-carbide and gallium-nitride transistors control electric cars, solar installations and wind turbines with efficiencies above 98 %.
- **Light-emitting diodes:** The blue LED (Akasaki, Amano and Nakamura, Nobel Prize 2014) made white LED lamps possible and has changed lighting throughout the world.

Further reading

- Paul Horowitz and Winfield Hill: *The Art of Electronics* (3rd ed., Cambridge 2015)
- Adel S. Sedra and Kenneth C. Smith: *Microelectronic Circuits* (8th ed., Oxford 2019)
- On this site: Circuit theory · Digital electronics · Solid-state physics